1.Title

Working Project Title

Network Intrusion Detection System

Team Details

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Program Track and Domain

Program Track	Domain
Science-Spoke	Machine Learning

Guide Details

Mentor Name: Sai Baba Basa
Designation: Master Trainer

2. Problem Statement

In the modern digital landscape, the rapid expansion of network traffic and the increasing sophistication of cyber-attacks have rendered traditional security measures inadequate. Conventional signature-based Intrusion Detection Systems (IDS) rely on predefined databases of known threats, making them largely ineffective against "zero-day" exploits and rapidly evolving malware variations. Furthermore, contemporary network data is characterized by high dimensionality and massive volume, often containing significant noise, redundant features, and severe class imbalances where malicious activities are dwarfed by benign traffic. There is a critical need for an automated, intelligent system capable of processing large-scale datasets like CICIDS2017 to distinguish between normal behavior and diverse attack vectors—such as DDoS, PortScan, and Web Attacks—with high precision and minimal false alarms.

3. Objectives

The primary goal of this project is to develop and evaluate a robust machine learning-based Network Intrusion Detection System. The specific objectives are as follows:

- **Data Pre-processing and Cleaning:** To handle large-scale network traffic data by addressing missing values, removing duplicates, and ensuring data consistency for reliable model training.
- **Exploratory Data Analysis (EDA):** To perform statistical analysis and visualization to identify traffic patterns, detect outliers, and understand the distribution of various attack classes.
- **Feature Selection and Dimensionality Reduction:** To utilize statistical methods and model-based techniques, such as Random Forest feature importance, to identify the most relevant network attributes and improve computational efficiency.
- **Algorithm Implementation:** To implement and train multiple machine learning models, including Decision Tree, Random Forest, and XGBoost, to classify network traffic effectively.
- **Performance Evaluation:** To rigorously assess the models using metrics such as Accuracy, Precision, Recall, and F1-score, ensuring the system can accurately detect intrusions while maintaining a low false-positive rate.
- **Comparative Analysis:** To compare the effectiveness of different algorithms to determine the most robust approach for real-world network security applications.

4. Methodology / Implementation Plan

The implementation of the Network Intrusion Detection System is executed through a systematic seven-phase pipeline, beginning with data acquisition and ending with a rigorous comparative analysis of machine learning models. Initially, the **CICIDS2017 dataset** is sourced and assessed to understand the various traffic categories, including both benign behavior and modern attack vectors. This is followed by a comprehensive **Data Pre-processing and Cleaning** phase, where the raw network logs are refined by removing null or infinite values, eliminating duplicate records, and scaling numerical features to ensure that no single attribute disproportionately influences the model's learning process.

Once the data is cleaned, **Exploratory Data Analysis (EDA)** is performed to visualize class distributions and identify correlations between different network features, such as flow duration and packet length. To enhance computational efficiency and model accuracy, a **Feature Selection** process is implemented using Random Forest importance scores, allowing the system to focus only on the most critical indicators of malicious activity. The optimized dataset is then split into an **80% training set and a 20% testing set** to facilitate unbiased evaluation. During the **Model Building** stage, multiple algorithms—including **Decision Tree, Random Forest, and XGBoost**—are trained and fine-tuned. The final phase involves a detailed **Performance Evaluation** using a confusion matrix and metrics such as precision, recall, and F1-score to determine which algorithm offers the most robust defense against contemporary network intrusions.

5. Resources and Timeline

A. Hardware & Software Requirements

Hardware Requirements

- **Processor:** Intel Core i5 or higher (or AMD equivalent) to support multi-threaded processing.
- **RAM:** Minimum 8 GB (16 GB recommended) to handle large datasets in memory.
- **Storage:** At least 50 GB of free space for the raw dataset, processed files, and model storage.
- **GPU (Optional):** An NVIDIA Graphics Card for faster training of complex models like XGBoost.

Software Requirements

- **Operating System:** Windows 10/11, macOS, or Linux.
- **Programming Language:** Python 3.8 or higher.
- **Platform:** Jupyter Notebook or Google Colab for coding and visualization.

Key Libraries:

- **Pandas & NumPy:** For data cleaning and mathematical operations.
- **Scikit-Learn:** For building Decision Tree and Random Forest models.
- **XGBoost:** For the high-performance gradient boosting model.
- **Matplotlib & Seaborn:** For creating graphs and heatmaps.
- **Pickle:** For saving the final trained models.

B. Dataset

The project uses the **CICIDS2017 dataset**, which is a widely recognized collection of modern network traffic data. It is preferred over older datasets because it contains contemporary traffic patterns and a wide variety of real-world cyber-attacks.

Key Highlights of the Dataset:

- **Modern Traffic:** It includes up-to-date network behaviors and realistic background traffic (like web browsing and email).
- **Size and Detail:** The dataset is large and complex, containing **80 different network features** (such as flow duration, packet size, and transmission speed).
- **Variety of Attacks:** It covers several common types of cyber-threats, including:
  - **DoS/DDoS:** Attacks that try to crash a system by overloading it.
  - **PortScan:** Probing a network to find open doors for an attack.
  - **Web Attacks:** Attempts to breach web applications or steal data.
  - **Botnets:** Networks of infected computers controlled by a hacker.
- **Data Split:** For this project, the data is divided into:
  - **Training Set (80%):** Used to teach the models how to recognize patterns.
  - **Testing Set (20%):** Used to check how accurately the models can identify new, unseen threats.

C. Project Timeline

Phase	Task	Tentative Completion Date
Phase 1	Topic selection and problem definition	Week 1
Phase 2	Dataset collection and study	Week 2
Phase 3	Data preprocessing and image preparation	Week 3
Phase 4	Model development and training	Week 4-5
Phase 5	Model evaluation and testing	Week 6
Phase 6	Result analysis and visualization	Week 7
Phase 7	Final report preparation and project submission	Week 8

6. Expected Outcomes / Conclusion

In conclusion, the development of the Network Intrusion Detection System (NIDS) confirms that machine learning is a highly effective approach for identifying modern cyber threats. By shifting from traditional signature-based methods to anomaly-based detection using the CICIDS2017 dataset, the system successfully identified a wide range of attacks, including DoS, PortScan, and Web Attacks. The project demonstrated that ensemble models like Random Forest and XGBoost provide superior performance, achieving high accuracy and low false-positive rates even with complex, high-dimensional data. Furthermore, the results highlight that rigorous data cleaning and feature selection are essential for creating a scalable and efficient defense framework. Overall, this project provides a robust foundation for building proactive security infrastructures capable of protecting data integrity in real-time digital environments.

7. References

- **GeeksforGeeks:** "Intrusion Detection System (IDS)" and "Machine Learning Algorithms" articles. Accessed for foundational concepts of network security, classification models, and Python implementation tutorials.
- **Canadian Institute for Cybersecurity (CIC):** "CICIDS2017 Dataset" documentation. Used for understanding network traffic features and attack categories.
- **Scikit-learn Documentation:** Technical guide for implementing **Decision Tree** and **Random Forest** classifiers, and performing data scaling and evaluation.
- **XGBoost Documentation:** Reference for the **Extreme Gradient Boosting** library, specifically for hyperparameter tuning and handling large-scale datasets.
- **Pandas & NumPy Documentation:** Official resources for data manipulation, handling missing values, and numerical processing of the CSV files.
- **Matplotlib & Seaborn:** Used as a reference for creating statistical charts, correlation heatmaps, and confusion matrices to visualize the results.